

Tax, Tip, Discount and Markup

Answer Key

Grade 7

Quick Answers

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|---|---|
| 1. (a) sales tax in dollars = 1.2, (b) total paid in dollars = 25.2 | in dollars = 49
6. 25 |
| 2. (a) tip in dollars = 7.2, (b) total paid in dollars = 47.2 | 7. 12.9
8. 70 |
| 3. (a) discount in dollars = 15, (b) sale price in dollars = 45 | 9. 80 |
| 4. 140.58 | 10. (a) sale price in dollars = 84, (b) sales tax in dollars = 6.72, (c) total paid in dollars = 90.72, |
| 5. (a) markup in dollars = 14, (b) selling price | — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

7.RP.A.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 17 tagged checks.

1. Book at 24 dollars, sales tax 5%. Find the tax and the total.

Solution: (a) The tax is a percent of the marked price: $\frac{5}{100} \times 24 = 1.20$ dollars. 1.20

(b) Adding the tax on multiplies the price by 1.05: $24 \times 1.05 = 25.20$ dollars, which is also $24 + 1.20$. 25.20

2. Meal at 40 dollars, 18% tip. Find the tip and the total.

Solution: (a) $\frac{18}{100} \times 40 = 7.20$ dollars. 7.20

(b) $40 \times 1.18 = 47.20$ dollars. 47.20

3. Jacket at 60 dollars, 25% off. Find the discount and the sale price.

- Solution:** (a) $\frac{25}{100} \times 60 = 15$ dollars off. 15
- (b) Taking 25% off leaves 75%, so $60 \times 0.75 = 45$ dollars, which is also $60 - 15$. 45
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4. Repair billed at 132 dollars, tax 6.5%. Find the total.

Solution: A 6.5% tax multiplies the bill by 1.065: $132 \times 1.065 = 140.58$ dollars. The tax alone is 8.58 dollars. 140.58

5. Lamp cost 35 dollars, marked up 40%. Find the markup and the selling price.

Solution: (a) The markup is a percent of what the shop *paid*: $\frac{40}{100} \times 35 = 14$ dollars. 14

(b) $35 \times 1.40 = 49$ dollars, which is also $35 + 14$. 49

6. Game was 48 dollars, now 36 dollars. What percent was taken off?

Solution: The amount taken off is $48 - 36 = 12$ dollars, and a discount percent is always measured against the *original* price: $\frac{12}{48} \times 100 = 25$ percent. 25

Common wrong answer: measuring the 12 dollars against the sale price gives $\frac{12}{36} \times 100 = 33.3$ percent.

7. Bill of 86 dollars, 15% tip. How much is the tip?

Solution: $\frac{15}{100} \times 86 = 12.90$ dollars. A quick check: 10% is 8.60 and half of that is 4.30, and $8.60 + 4.30 = 12.90$. 12.90

8. Chair sells for 84 dollars after a 20% markup. What did the shop pay?

Solution: The selling price is the cost multiplied by 1.20, so the cost is the selling price divided by 1.20: $84 \div 1.20 = 70$ dollars. Check: 20% of 70 is 14, and $70 + 14 = 84$. 70

Common wrong answer: taking 20% off the 84 dollars gives 67.20 dollars, and marking that up again would not return to 84.

9. Coat on sale for 68 dollars after 15% off. What was the original price?

Solution: A 15% discount leaves 85% of the original, so $0.85x = 68$ and $x = 68 \div 0.85 = 80$ dollars. Check: 15% of 80 is 12, and $80 - 12 = 68$. 80

Common wrong answer: adding 15% of the sale price back on gives $68 \times 1.15 = 78.20$ dollars, which is not the original.

10. Tent marked 120 dollars, 30% off, then 8% tax on the sale price.

Solution: (a) Taking 30% off leaves 70%: $120 \times 0.70 = 84$ dollars.

84

(b) The tax is charged on the sale price, not the marked price: $\frac{8}{100} \times 84 = 6.72$ dollars.

6.72

(c) $84 + 6.72 = 90.72$ dollars, which is also $120 \times 0.70 \times 1.08$.

90.72

(d) **Instructor-judged.** No — the total does not change. Taking the discount multiplies by 0.70 and adding the tax multiplies by 1.08, and multiplying by two factors gives the same product in either order. Taxing first gives $120 \times 1.08 = 129.60$ dollars, and 30% off that is $129.60 \times 0.70 = 90.72$ dollars, the same total. Full credit needs the reason to be about multiplying by the two factors — either named as a property or shown by working the second order through.
